

Prototyping and modeling of the centrifugal casting process for paraffin waxes

Davide Masato, Marco Sorgato & Giovanni Lucchetta

To cite this article: Davide Masato, Marco Sorgato & Giovanni Lucchetta (2017) Prototyping and modeling of the centrifugal casting process for paraffin waxes, Materials and Manufacturing Processes, 32:16, 1823-1830, DOI: [10.1080/10426914.2017.1317791](https://doi.org/10.1080/10426914.2017.1317791)

To link to this article: <https://doi.org/10.1080/10426914.2017.1317791>



Published online: 16 May 2017.



Submit your article to this journal [↗](#)



Article views: 642



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 2 View citing articles [↗](#)



Prototyping and modeling of the centrifugal casting process for paraffin waxes

Davide Masato , Marco Sorgato , and Giovanni Lucchetta 

Department of Industrial Engineering, University of Padova, Padova, Italy

ABSTRACT

Centrifugal casting is a technology used for manufacturing hybrid rocket paraffin grains. This technology helps avoiding voids formation inside the solid paraffin as it cools. Voids are formed because of air bubbles being entrapped while pouring and because the liquid wax shrinks by 17–19% upon cooling. In this work, the centrifugal casting process for the manufacturing of paraffin cylinders was prototyped at two different scales considering critical casting issues. The effects of process parameters (rotational speed, melt temperature, and flow rate) on the tensile properties of the manufactured grains were analyzed. The results of the optimization conducted at the lower scale (2.5 kg) were up scaled to manufacture 25 kg grains. The resulting mechanical properties complied with the design specifications, and they were better than those characterized from the gravity cast wax. A numerical model of growth and dissolution of bubbles during the process was then developed to predict the quality of the castings. The numerical results showed how increasing the mold rotational speed up to 1800 rpm reduced the removal time. However, compared to grains solidification time, the predicted removal times were much shorter, proving the advantage of centrifugal casting in counteracting voids formation.

ARTICLE HISTORY

Received 17 October 2016
Accepted 8 March 2017

KEYWORDS

Casting; centrifugal;
characterization; engine;
hybrid; manufacturing;
optimization; paraffin;
process; rocket

Introduction

Chemical fuels for aerospace rocket applications can be divided into three different classes, according to the phase the propellant is stored in: liquid, solid, and hybrid. In a liquid rocket, a liquid or gaseous oxidizer and a fuel are contained into two different tanks and then injected in the combustion chamber. In a solid rocket, fuel and oxidizer are mixed together in a solid matrix, which is directly stored in the combustion chamber. In hybrid rockets, one component, usually the oxidizer, is stored as a liquid or as a gas, while the other (i.e., the fuel) is solid and it is located inside the combustion chamber.

Hybrid propulsion combines the advantages of liquid and solid propulsion, thus having the potential to replace current rocket motors functionality as boost phase launch vehicle components [1]. Hybrid rocket engines (HRE) are well suited to applications requiring throttling, command shutdown, and restart. Moreover, they are nontoxic propellants, which are suitable for operations (manufacturing and launch) that would benefit from a non-self-deflagrating propulsion system [2]. Such applications include primary boost propulsion for space launch vehicles and upper stages [3]. However, despite all these advantages and the high performances in terms of specific impulse, the development and the application of hybrid systems have been delayed by the low regression rate of the fuel [4].

Currently, the most used fuel for HRE is hydroxyl-terminated polybutadiene (HTPB), which has a low cost, it is easy to process, and it does not self-deflagrate under any condition [5]. However, HRE made of HTPB need a complex

design to guarantee the required fuel surface area, leading to manufacturing complexity and possible structural failures [6].

In the last decade, some researches highlighted the marked regression rate increase offered by solid paraffins (waxes) with respect to conventional polymeric fuels [7]. Paraffin waxes (C_nH_{2n+2}) are thermoplastic polymers made of alkanes with carbon number (n) between 25 and 60. In general, they are cheap materials used for low-added-value products, such as candles. However, in hybrid propulsion applications, they offer significant advantages because of their high burning rate. Indeed, due to the so-called entrainment mass transfer mechanism, which consists in the mechanical entrainment of liquid droplets in the oxidizer mass flow, they can burn up to three times faster than HTPB fuels [8].

Paraffin waxes available in the market are characterized by broadly different thermal, mechanical, and rheological properties; therefore, it is necessary to identify and select those that can suit HRE applications. Moreover, their extensive use is still hindered by the lack of suitable and reliable manufacturing technology. Indeed, paraffin-based HRE grains need to sustain launch loads and stresses. Satisfactory mechanical properties can be achieved by pursuing different strengthening strategies, such as material selection [9] and process optimization [10].

The centrifugal casting process has been applied to HRE manufacturing for its clear advantages of reducing cracks and void formation inside the solid paraffin wax [11]. In fact, paraffin waxes shrink upon solidification by 17–19%, depending on the grade of the wax, thus leading to the formation of defects that can compromise the structural integrity of the castings. The process is characterized by a permanent

axisymmetric mold that is continuously rotated about its longitudinal axis at high speed as the molten polymer is poured [12]. The molten paraffin is centrifugally thrown toward the mold wall, where it solidifies after cooling.

In general, castings for HRE applications are thick-wall cylinders with an inner circular port. Due to their typical mass (e.g., 10^1 – 10^2 kg), HRE paraffin grains are difficult to produce by gravity casting. In fact, due to the low thermal conductivity and the high heat capacity, this process usually yields internal cracks and inhomogeneity. Moreover, paraffin waxes shrink so much upon cooling that it is not possible to manufacture a thick-wall cylinder of regular geometry without any internal defect, and this would eventually compromise their mechanical resistance. Centrifugal casting was therefore selected over gravity casting in order to improve both manufacturability and grains mechanical properties.

In the centrifugal casting process, the centrifugal action exerted by the rotation favors the adhesion of the wax to the inner wall of the mold, allowing the manufacturing of a hollow product. The directional solidification leads to a reduction in the formation of shrinkage porosities and promotes the buoyancy of entrapped air bubbles toward the innermost layer of the cylinder, which can then be trimmed.

In the design of the centrifugal casting process, the main phenomena that have to be considered are related to the formation of different defects within the solidified paraffin wax, such as segregation banding, radial and circumferential cracks, longitudinal hot tears, and internal stresses. Moreover, another manufacturing issue that has to be taken in consideration is the so-called raining phenomenon [13]. In fact, at a low spin speed or high pouring rate, the centrifugal force is lower than the gravitational one, and as a consequence the melt loses contact with the mold. This is particularly critical for paraffin waxes, whose low viscosity reduces the drag exerted by the mold, hindering the melt to reach the rotational velocity that is necessary to overcome the force of gravity. The raining phenomenon, which can lead to the onset of detrimental vibrations, can be controlled by defining the proper selection of process parameters, in particular flow rate, mold rotational speed and melt temperature [14].

The low viscosity of the molten wax also favors the entrapment of air bubbles during mold filling, leading to the eventual formation of internal porosities in the castings. These defects are then eliminated during mold spinning by the action of the centrifugal force, which can help their dissolution or movement toward the innermost layer of the grain [15].

To date, very few studies in the literature have reported the application of centrifugal casting to the manufacturing of HRE. Moreover, this manufacturing technology has not been fully investigated and verified, especially in relation to base materials and grains properties. In this work, the process technology for the manufacturing of paraffin-based HRE was experimentally prototyped and optimized. The application of centrifugal casting to the manufacturing of HRE was evaluated considering the optimization of grains mechanical properties, setting the basis for the development of new hybrid aerospace applications.

A suitable material was identified among three commercial paraffin waxes considering their mechanical, rheological, and

thermal properties. The design and the optimization of the process were conducted optimizing the mechanical properties of the cast grains. The effects of controllable process parameters were systematically investigated on a prototypal scale, and then, the results were transferred to the real scale and verified.

A numerical model, which combines the growth and the dissolution of air bubbles during the centrifugal casting process, was developed and applied to the prediction of the quality of the casting, in terms of internal porosities. The numerical solution of the model, which captures the combined effects of buoyancy, drag, and diffusion, allowed the prediction of the time required by bubbles either to dissolve into the molten polymer or to reach the innermost layer of the grain.

Materials and Methods

The hybrid rocket engine considered in this study was designed to produce a maximum thrust of about 7 kN. In agreement with this specification, the following dimensions of the HRE grain were defined:

- length: 400 mm;
- internal diameter: 70 mm;
- external diameter: 310 mm.

Thus, the manufactured single-port grains were characterized by a weight of about 25 kg and a ratio between the external and internal diameters of 4.4.

The functionality of the designed HRE is strongly related to the properties that the wax should have during HRE use. In particular, the thermo-mechanical loads that most critically stress the grains are the oxidizing internal pressure (up to 20 bar) and the environmental temperature variation (between -10 and 50°C). A finite element structural analysis of the designed grain was performed in a previous study, and these design specifications were translated into the following material property requirements:

- minimum stress at break (at room temperature): 3 MPa;
- minimum melt temperature: 50°C .

Three commercial pure paraffin waxes were selected as candidate fuels for the manufacturing of HRE grains: Paraffin A, Paraffin B, and Paraffin C. All of them are microcrystalline waxes, characterized by a higher percentage of branched molecules, which increases the molecular weight.

The thermal properties of the pure waxes were characterized by means of differential scanning calorimetry (TA Instruments, DSC Q200, New Castle, DE, USA), in agreement with ISO 11357–3. All tests were performed from -80 to 150°C with a temperature ramp of $10^\circ\text{C}/\text{min}$, using nitrogen gas ($50\text{ mL}/\text{min}$) as heating/cooling gas.

Rheological characterizations were performed using a parallel plate (diameter 50 mm, gap 0.5 mm) rheometer (TA Instruments, ARES), and tests were performed by varying the shear rate from 5 to 2000 s^{-1} , at a constant temperature of 120°C .

The mechanical behavior of the paraffins was characterized by means of uniaxial tensile tests at room temperature using a speed of $1\text{ mm}/\text{s}$, in accordance with ISO 527–2:2012. Small specimens (type 1BA) were obtained by gravity casting the molten waxes into a dumbbell shaped cavity in a rapid prototyped mold. A thermally controlled procedure was adopted in

order to prevent the formation of internal defects due to shrinkage. Elastic modulus, stress at break, and strain at break were taken in consideration for the comparison of the mechanical properties of the tested paraffins.

A specifically designed machine was built to perform centrifugal casting tests with paraffins. The machine, shown in Fig. 1, was characterized as follows:

- composite mold (carbon fiber T300 – epoxy resin) and sealing plugs to control the thermal boundary conditions of the process and promote a directional inwards solidification of the wax;
- compatibility with a smaller prototypal mold, for casting 2.5 kg paraffin grains (length: 180 mm; internal diameter: 50 mm; external diameter: 140 mm), which was used to investigate the effect of controllable process parameters and then upscale the process to the 25 kg grains;
- control of the mold rotational speed using a Labview (National Instruments, Austin, TX, USA) application and temperature monitoring of the poured wax by means of a temperature probe (Hasco, Z2511/Typ, Römerweg, Austria).

The capabilities of the centrifugal casting process to produce paraffin grains characterized by adequate structural properties were at first investigated using the prototypal scale version of the machine (2.5 kg wax grains) and then confirmed on the 25 kg scale.

In order to determine a range of values that would suit the 25 kg scale, different values of the rotational speed were tested at the prototypal scale. Controlling both machine vibrations and the raining phenomenon, the speed was increased with steps of 50 rpm. The lower limit, below which the paraffin melt was not accelerated enough to overcome the gravitational force, was identified at 800 rpm. The upper limit was found at 1800 rpm, which corresponds to machine maximum rotational speed, as no critical vibrations were observed.

A design of experiments (DoE) approach was followed to study the effect of controllable centrifugal casting parameters on the mechanical properties of the manufactured grains. A two-level full factorial plan was designed considering the following factors: mold rotational speed, paraffin melt temperature, and volumetric flow rate. The upper and lower levels for the factors were selected considering technological limits of the available experimental setup (maximum rotational speed: 1800 rpm), issues related to machine vibrations (maximum volumetric flow rate: 80 cm³/s), and rheological

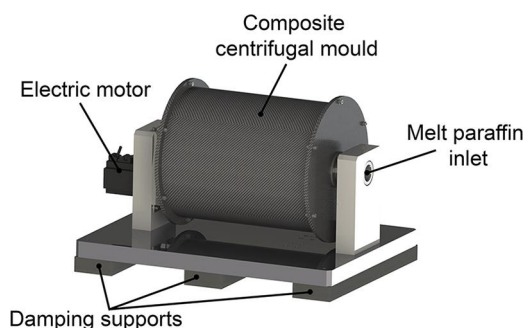


Figure 1. Centrifugal casting machine design and main components.

Table 1. Process parameters setting for the experimental analysis.

Factor	Level	
	(-)	(+)
Rotational Speed (rpm)	800	1800
Melt Temperature (°C)	80	115
Volumetric Flow Rate (cm ³ /s)	8	80

properties of the paraffins melt. The levels for the investigated parameters are reported in Table 1.

The mechanical properties of the grains were chosen as the response variable for the analysis. Both stress and deformation at break were analyzed by means of uniaxial tensile strength tests, which were performed on specimens machined from an intermediate radial position, according to ISO 527. Each run of the DoE experimental plan was replicated three times in a completely randomized order, thus producing 24 prototypal scale grains. All the grains were manufactured using new material taken from the same batch. For each grain, three tensile specimens were machined and then tested.

The information obtained from the experimental investigation on the prototypal scale was then transferred to the manufacturing of the 25 kg paraffin grains. The optimum selection of process parameters was set accordingly, and the mechanical properties of the large-scale grains were then verified in two different radial positions, at 10 mm from the inner grain surface (internal position) and at 10 mm from the external surface (external position).

Results and Discussion

The tested paraffins exhibited a common thermal behavior (Fig. 2), typical of microcrystalline waxes, characterized by a main melting peak and a lower temperature transition peak [16]. However, Paraffin B and C showed a wider melting changing phase interval with more or less distinctive peaks and a lower specific heat flow, hence indicating that they are characterized by the presence of a secondary phase, that is, iso-paraffins.

The results of the rheological measurements indicated that all the considered waxes behave as Newtonian fluids, as shown in Fig. 3. Table 2 reports the viscosity for the three polymers, indicating that Paraffin A has the lowest viscosity; hence, it likely has the lowest regression rate.

The tensile tests, performed on gravity cast specimens, indicated that the three waxes are characterized by different

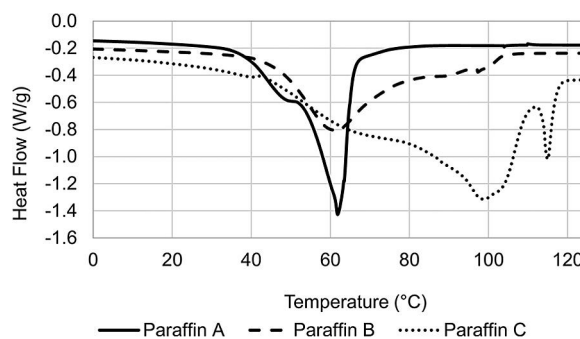


Figure 2. DSC melting curves for the three paraffins.

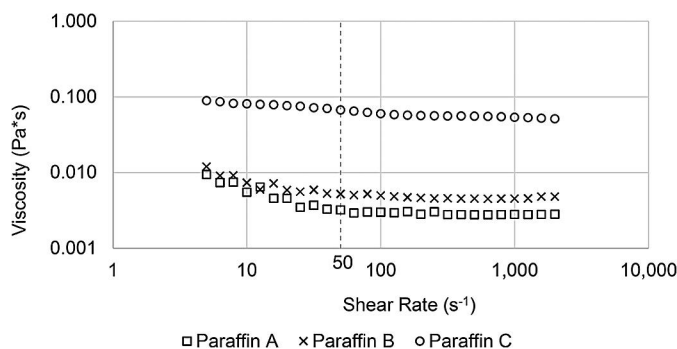


Figure 3. Rheological characterization for the three paraffins, exhibiting their Newtonian behavior.

Table 2. Summary of the main characterized properties for the three selected commercial paraffins.

Material	Paraffin A		Paraffin B		Paraffin C	
	Avg. Val.	Std. Dev.	Avg. Val.	Std. Dev.	Avg. Val.	Std. Dev.
Property						
Viscosity ($\dot{\gamma} = 50 \text{ s}^{-1}$) (Pa·s)	0.0032	0.0004	0.0052	0.0006	0.0672	0.0004
Elastic Modulus (MPa)	571	30	528	43	427	46
Stress at break (MPa)	3.11	0.07	3.20	0.16	4.14	0.37
Strain at break	0.023	0.005	0.020	0.002	0.032	0.008

mechanical properties, as shown in Fig. 4. In particular, Paraffin C exhibited the higher mechanical resistance having an average stress at break 36% higher than Paraffin A, which had the smallest resistance. The values of the deformation at break are comparable considering the standard deviation of the measured values.

Considering the results of the characterization, Paraffin A was selected and used for the centrifugal casting of HRE grains, because its mechanical properties suit the design requirements and it has the lowest viscosity (i.e., the highest regression rate).

The process influence on the mechanical properties of centrifugal casting prototypal scale grains (2.5 kg) was experimentally investigated both in terms of stress and deformation at break, which were then compared with the basic properties of Paraffin A, as shown in Table 3.

The characterization of the prototypal scale grains indicated that different selections of process parameters significantly

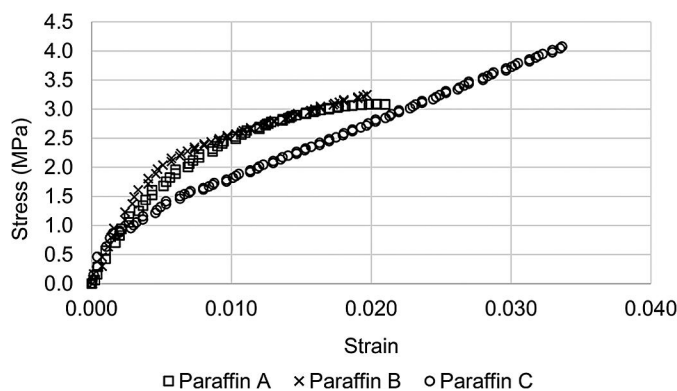


Figure 4. Tensile stress tests for the three paraffins.

Table 3. Effect of centrifugal casting process parameters on paraffin grains mechanical properties.

Test	Rotational Speed (rpm)	Melt Temperature (°C)	Volumetric Flow Rate (cm³/s)	σ_r (MPa)		ϵ_r	
				Avg. Val.	Std. Dev.	Avg. Val.	Std. Dev.
Paraffin A	/	/	/	3.12	0.35	0.023	0.005
a	1800	80	80	3.29	0.15	0.042	0.005
b	1800	115	8	3.15	0.12	0.043	0.005
c	800	80	80	3.08	0.54	0.037	0.008
d	800	115	80	2.86	0.16	0.039	0.003
e	800	115	8	2.72	0.26	0.044	0.007
f	800	80	8	2.91	0.28	0.028	0.006
g	1800	115	80	3.33	0.21	0.033	0.007
h	1800	80	8	3.28	0.23	0.038	0.006

Table 4. ANOVA table for the experimental plan.

Factor	p value	
	Stress at break	Deformation at break
n	0.004	0.635
T_m	0.296	0.479
t_c	0.288	0.916
$n \cdot T_m$	0.482	0.273
$n \cdot t_c$	0.797	0.617
$T_m \cdot t_c$	0.757	0.186

affect the mechanical properties of the centrifugal cast wax. Indeed, for some combinations of process parameters, the structural requirements were not satisfied, showing the importance of selecting the adequate values of the centrifugal casting parameters.

The analysis of variance (ANOVA) performed on the results of the designed experimental plan indicated that, among the investigated factors, the rotational speed is the only statistically significant parameter affecting the stress at break (p value = 0.004), as reported in Table 4. Indeed, the main effect of increasing the rotational speed from 800 to 1800 rpm is to increase the stress at break by +13% (Fig. 5). Conversely, the effects of the melt temperature and the volumetric flow rate were not significant (p value > 0.05). Observing the deformation at break, no significant main effect was observed (p value > 0.05), due to the high dispersion of the measured elongation values (Fig. 6).

Comparing the mechanical properties between centrifugal cast grains and gravity cast waxes, it was observed that the former has always a higher deformation at break, while they have a higher value of the stress at break only when cast at high speed.

The centrifugal casting process was up scaled for the manufacturing of 25 kg HRE grains, considering the information gathered from the experimentation at the prototypal scale and some scaling issues. In particular, the following problems were addressed in order to set the process parameters:

- A volumetric flow rate of $8 \text{ cm}^3/\text{s}$ was set to control the vibrations of the machine. Indeed, at high values of the flow rate, the molten paraffin is not able to distribute itself uniformly along the centrifugal mold, causing the onset of detrimental instabilities during rotation.
- An external forced-air cooling system was used to improve the mold heat dissipation and decrease the process cycle

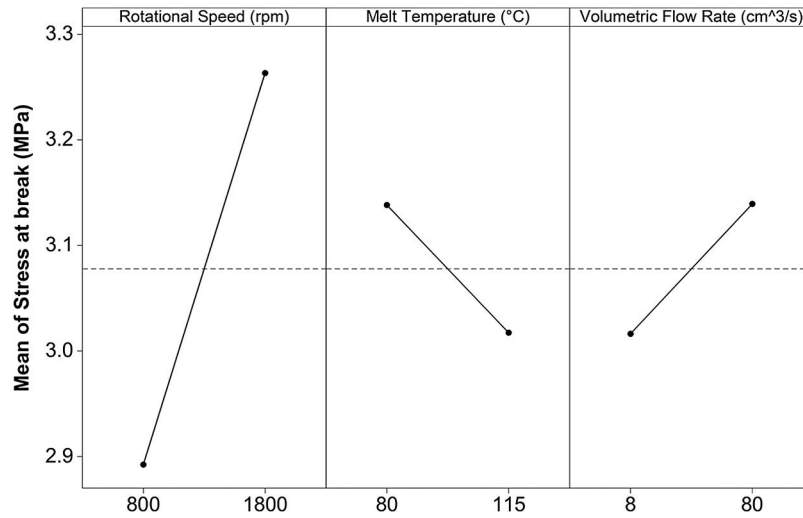


Figure 5. Main effect plots for the stress at break.

time. Moreover, an internal forced-air cooling system was used to increase heat dissipation also from the inner surface of the grain.

- The raining effect was controlled defining a G factor as the ratio between gravitational F_g and centrifugal F_c forces:

$$G = F_g / F_c = \frac{mr\omega^2}{mg} = \left(\frac{r}{g}\right) \left(\frac{\pi}{30}\right)^2 n^2 \quad (1)$$

$$n = 29.9 \left(\frac{G}{r}\right)^{0.5}, \quad (2)$$

where m is the grain mass, r is the mold radius, ω is the angular velocity, and n is the mold rotational speed. From the maximum and minimum values of the rotational speed, determined with the experiments carried out at the prototypal scale, the following range for the G factor was defined:

$$540 \text{ rpm} < n < 1200 \text{ rpm} \quad (3)$$

Preliminary centrifugal casting tests indicated that the highest rotational speed (i.e., 1200 rpm) led to the rotational-vibrational

coupling of the centrifugal casting system. Thus, being the rotational frequency close or identical to a natural internal vibration frequency of the machine, it was necessary to decrease the upper limit of the range. Consequently, considering its positive effect on the mechanical properties and in order to avoid detrimental vibrations during operations, the mold rotational speed was set at 900 rpm. The melt temperature was set at an intermediate value of 100°C, as this factor was reported to be statistically not significant.

The scalability of the process was verified producing three 25 kg centrifugal cast grains, according to the defined process parameters. Each of the manufactured grain was characterized by sampling three tensile specimens along the grains length in two different radial positions (internal and external), at 10 mm from the grain inner and outer surfaces, respectively.

The results of the mechanical characterization (Table 5) indicated that the structural properties of the manufactured 25 kg paraffin grains complied with the defined requirements, thus indicating the soundness of process parameters selection. Indeed, the values of the stress at break obtained for the 25 kg

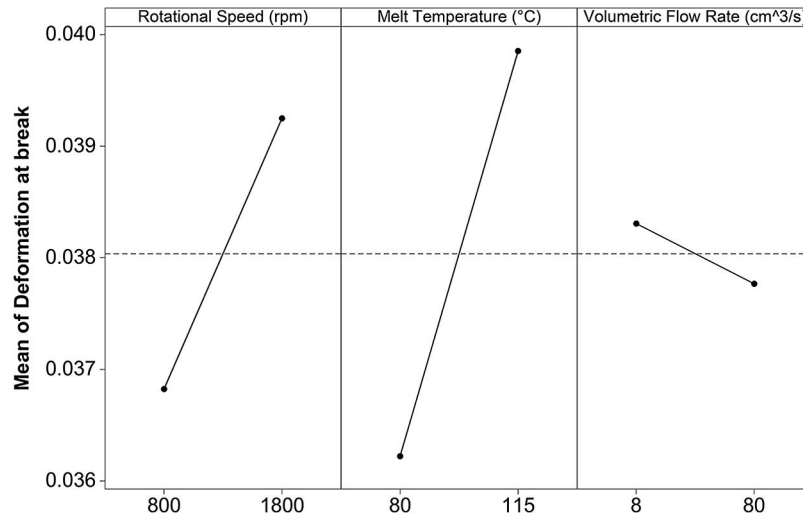


Figure 6. Main effect plots for the deformation at break.

Table 5. Mechanical properties of the 25 kg centrifugal casting grains.

Position	σ_r (MPa)		ϵ_r	
	Avg. Val.	Std. Dev.	Avg. Val.	Std. Dev.
Internal	3.21	0.23	0.038	0.015
External	3.30	0.22	0.040	0.008

scale grains are comparable to those obtained for the prototypal scale with a higher rotational speed (i.e., 1800 rpm). Thus, indicating the positive effect of the increased scale on the mechanical properties, due to the higher paraffin melt mass that causes longer solidification times, allowing the higher rotational speed to be more effective.

The mechanical properties of tensile bars sampled in different radial position were not significantly different, as shown in Fig. 7. Hence indicating that good homogeneity and the absence of main defects were achieved in the radial direction.

The main defects that could result from the centrifugal casting of paraffin grains are internal shrinkage porosities and air bubbles. In particular, the latter, which are formed when the molten wax is poured into the mold, can negatively affect the final mechanical properties of the solid paraffin grain.

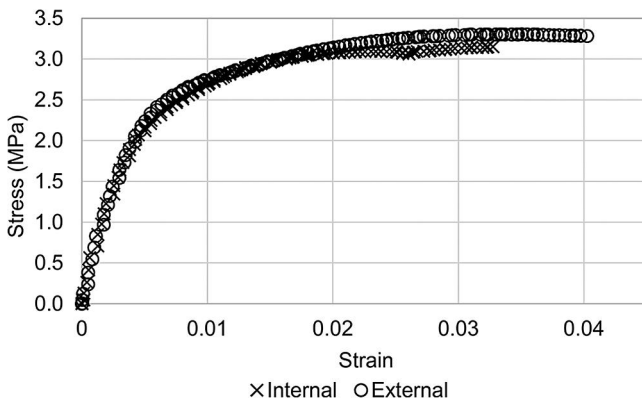
Centrifugal casting is a recognized method for eliminating bubbles in polymer castings [14]. In particular, there are two mechanisms contributing to the elimination of internal bubbles:

- diffusion, which occurs when the concentration of gas dissolved within the bubble wall is greater than the concentration of gas in the surrounding solution;
- buoyant transport, that is, the bubbles tend to move toward the center axis of rotation.

The diffusion mechanism can be quantified by the rate of change of bubble radius, which can be derived from the work of Epstein and Plesset [17]:

$$\frac{dR}{dt} = \frac{k(c_{S\infty}(c_f - 1) - \frac{2k_d\sigma}{R})}{\rho_\infty + \frac{2\epsilon}{3R}} \left(\frac{1}{R} + \frac{1}{\sqrt{\pi kt}} \right) \quad (4)$$

where $\tau = \frac{2M\sigma}{RT}$, M is the molecular weight of the gas, B the universal gas constant (8.314 J/(mol K)), k the coefficient of diffusion, ρ_∞ the density of the gas, and $c_{S\infty}$, c_f are concentration parameters.

**Figure 7.** Mechanical properties of tensile specimens machined in internal and external radial positions.**Table 6.** Parameters and coefficient used for the numerical model.

Parameter	Unit	Value
Bubble initial position	m	0.155
Bubble exit position	m	0.035
Surface tension	N/m	0.02
Paraffin melt density	kg/m ³	760
Diffusion coefficient	m ² /s	1.47*10 ⁻⁹
Initial air concentration	–	1.05
Henry's coefficient	kg(air)/m ³ (paraffin)/Pa	8.96*10 ⁻⁸

The radial velocity v_r of a bubble that moves within the melt polymer is obtained balancing the buoyant and the drag forces:

$$v_r = \frac{(\rho_a - \rho_f)}{12\mu_f} r\omega^2 d^2 \quad (5)$$

where ω is the angular velocity of the centrifuge, d is the initial diameter of the bubble, r the initial distance of the bubble from the axis of rotation, ρ_a is the density of the air within the bubble, and ρ_f , μ_f the density and the viscosity of the paraffin melt.

The bubbles removal from the paraffin melt was simulated considering both the change in their radius and their transport in the radial direction. The final quality of the paraffin grain was then predicted using a numerical solution of first-order nonlinear Eqs. (4) and (5), which were solved using Matlab[®] (Natick, MA, USA). Table 6 reports the value of the parameters and of the coefficients used to solve the numerical model.

The time required for bubbles to either dissolve into solution or leave the region of interest, at different processing conditions, was estimated considering an isothermal process in which the mold is completely filled before spinning and the bubbles are initially at the external diameter. Different values for the initial diameter of air bubbles were taken into account in order to observe its relation with bubble dynamics.

The simulation of the dynamics of air bubbles trapped during mold filling indicated that the complete removal of bubbles from the molten wax occurs in times ranging from less than a second to less than a minute (Table 7). Considering the 25 kg grains solidification time, which is about 8 hours, it was then observed that bubbles introduced while pouring the paraffin melt are removed well before the complete solidification. In particular, the maximum removal time observed was 40 s, for a rotational speed that was below the experimental range.

Moreover, it was observed that the initial diameter of the bubbles is responsible for the different elimination mechanism. Indeed, diffusion dominates for smaller bubbles, which get even smaller until they dissolve into solution. Conversely, bigger bubbles are subjected to an increase in their diameter until they reach the innermost layer due to the drag mechanism.

The model allowed the evaluation of the effect of controllable process parameters. In particular, a positive effect of higher rotational speed on the removal time was observed. The effect of this parameter is marked at low speed values,

Table 7. Time required for bubbles removal during spinning as a function of different initial defects dimensions and process parameters.

Initial Dimensions (μm)		Melt Temperature: 80°C				Melt Temperature: 115°C			
		10	95	300	1000	10	95	300	1000
Rotational Speed (rpm)	250	0.32	39.79	25.42	2.84	0.30	11.38	0.93	0.10
	850	0.06	5.99	2.55	0.27	0.06	1.27	0.09	0.01
	1800	0.03	1.40	0.75	0.08	0.02	0.39	0.03	0.01
	2000	0.02	1.22	0.67	0.07	0.02	0.35	0.03	0.01

while it stabilizes above 1800 rpm. Moreover, the melt casting temperature affects the quality of the paraffin grains by reducing the time required for bubbles removal. Indeed, at higher temperature, the viscosity is lower, facilitating the movement of the bubbles within the molten paraffin.

Conclusions

The main objectives of this research were the prototyping and the characterization of the centrifugal casting process for the manufacturing of paraffin-based hybrid rocket engines.

Three commercial waxes were characterized aiming at identifying the most suitable for HRE application. The thermal, rheological, and mechanical properties were characterized, and Paraffin A was selected because of its potential performance during both launch and combustion. In particular, its very low viscosity is likely to give the highest regression rate.

The manufacturing of paraffin grains was then experimentally tested on a prototypal scale (2.5 kg grains) using a specifically designed centrifugal casting machine. A procedure to sample small tensile specimens from the manufactured grains and to characterize their mechanical properties was established and evaluated. The investigation of the effects of controllable process parameters indicated that rotational speed is the most important factor affecting the mechanical properties of the grains (stress at break increased by +13% at high speed). Conversely, the effects of the other parameters (i.e., paraffin casting temperature and volumetric flow rate) were reported to be statistically negligible.

The process was then scaled up to the manufacturing of 25 kg grains. The rotational speed was defined considering the effect of the increased weight and dimensions on the ratio between the gravitational and centrifugal forces, which controls the raining phenomenon and eventually the final quality of the castings. The comparison with the tensile properties obtained by sound gravity casting specimens indicated that the centrifugal casting process can be effectively exploited for the manufacturing of paraffin-based HRE grains, satisfying the design requirements that were specified for a maximum thrust of 7 kN.

The process was then modeled in order to predict the time required for possible internal porosities to either dissolve into the molten paraffin or exit the paraffin melt during spinning. The simulation allowed the evaluation of the effects of different process parameters, considering wider variation ranges than the experimental ones. The numerical results showed how increasing the mold rotational speed reduced the removal time. However, the trend reached stabilization at about 1800 rpm and beyond that no significant improvements were

observed. Moreover, compared to grains solidification time, the predicted removal times were not relevant.

The experimental approach proposed in this paper, which is based on upscaling results from smaller centrifugal cast grains, has eventually been verified to be suitable for the manufacturing process design of large paraffin grains.

ORCID

Davide Masato  <http://orcid.org/0000-0002-9640-8179>

Marco Sorgato  <http://orcid.org/0000-0001-7293-0547>

Giovanni Lucchetta  <http://orcid.org/0000-0003-1421-8636>

References

- [1] Pastrone, D. Approaches to low fuel regression rate in hybrid rocket engines. *International Journal of Aerospace Engineering* **2012**, 2012, 1–12. doi:10.1155/2012/649753.
- [2] Jansen, D.; Kletzkine, P. Preliminary design for a 3 kN hybrid propellant engine. *ESA Journal* **1988**, 12, 421–439.
- [3] Davydenko, N.A.; Gollender, R.G.; Gubertov, A.M.; Mironov, V.V.; Volkov, N.N. Hybrid rocket engines: the benefits and prospects. *Aerospace Science Technology* **2007**, 11 (1), 55–60. doi:10.1016/j.ast.2006.08.008.
- [4] Chang, S.Y.; Lee, C.; Shin, K.S. The enhancement of regression rate of hybrid rocket fuel by various methods. In *43rd AIAA – Aerospace Sciences Meeting and Exhibit*, Reno, Nevada, January 10–13, 2005. doi:10.2514/6.2005-359.
- [5] Chiaverini, M.J.; Serin, N.; Johnson, D.K.; Lu, Y.; Kuo, K.K.; Risha, G.A. Regression rate behaviour of hybrid rocket solid fuels. *Journal of Propulsion and Power* **2000**, 16 (1), 125–132. doi:10.2514/2.5541.
- [6] Sutton, G.P.; Biblarz, O. *Rocket propulsion elements*; John Wiley & Sons: Hoboken, NJ, 2010, p. 784.
- [7] Karabeyoglu, M.A.; Altman, D.; Cantwell, B.J. Combustion of liquefying hybrid propellants: Part 1, general theory. *Journal of Propulsion and Power* **2002**, 18 (3), 610–620. doi:10.2514/2.5975.
- [8] Karabeyoglu, M.A.; Cantwell, B.J. Combustion of liquefying hybrid propellants: Part 2, Stability of liquid films. *Journal of Propulsion and Power* **2002**, 18 (3), 621–630. doi:10.2514/2.5976.
- [9] Boiocchi, M.; Merotto, L.; Maggi, F.; Di Landro, L.; Galfetti, L. Rheological and mechanical characterization of hybrid rocket solid fuels. In *European Conference For Aerospace Sciences*, Saint Petersburg, Russia, July, 2011.
- [10] DeSain, J.D.; Brady, B.B.; Metzler, K.M.; Curtiss, T.J.; Albright, T.V. Tensile tests of paraffin wax for hybrid rocket fuel grains. In *45th AIAA/ASME/SAE/ASEE Joint Propulsion Conference & Exhibit*, Denver, Colorado, August 2–5, 2009. doi:10.2514/6.2009-5115.
- [11] Karabeyoglu, M.A.; Zilliac, G.; Cantwell, B.J.; De Zilwa, S.; Castellucci, P. Scale-up tests of high regression rate liquefying hybrid rocket fuels. In *41st Aerospace Sciences Meeting and Exhibit*, Reno, Nevada, January 6–9, 2003. doi:10.2514/6.2003-1162.
- [12] Boros, C.; Konecny, P. Development of wax fuel grain for hybrid rocket motor. *Advances in Military Technology*, **2009**, 4 (2), 5–11.
- [13] Wei, S.; Lampman, S. *Centrifugal casting*; ASM Handbook, 2008, pp. 667–673. doi:10.1361/asmhba0005257.

- [14] Prasad, K.K.; Murali, M.S.; Mukunda, P.G. Analysis of fluid flow in centrifugal casting. *Frontiers of Materials Science in China* **2010**, *4* (1), 103–110. doi:[10.1007/s11706-010-0005-4](https://doi.org/10.1007/s11706-010-0005-4).
- [15] Mazzeo, A.D.; Lustrino, M.E.; Hardt, D.E. Bubble removal in centrifugal casting: combined effects of buoyancy and diffusion. *Polymer Engineering & Science* **2012**, *52* (1), 80–90. doi:[10.1002/pen.22049](https://doi.org/10.1002/pen.22049).
- [16] Meyer, G.; Thermal properties of micro-crystalline waxes in dependence on the degree of deoiling. *SOFW Journal*, **2009**, *135* (8), 43–50.
- [17] Epstein, C.E.P.; Plesset, M.S. On the stability of gas bubbles in liquid-gas solutions. *The Journal of Chemical Physics* **1950**, *18* (11), 1505–1509. doi:[10.1063/1.1747520](https://doi.org/10.1063/1.1747520).